

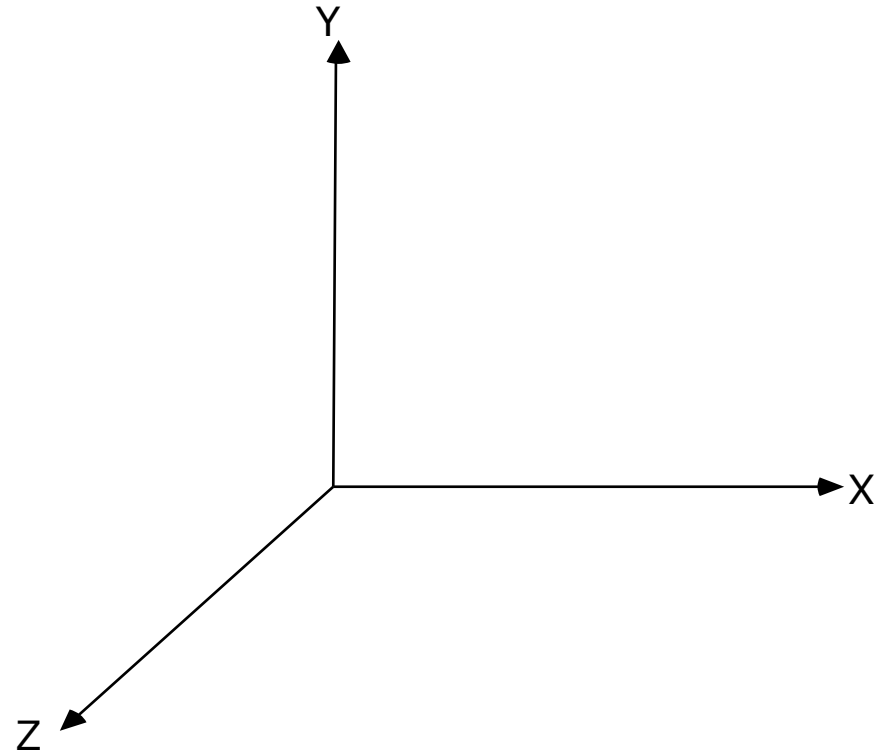
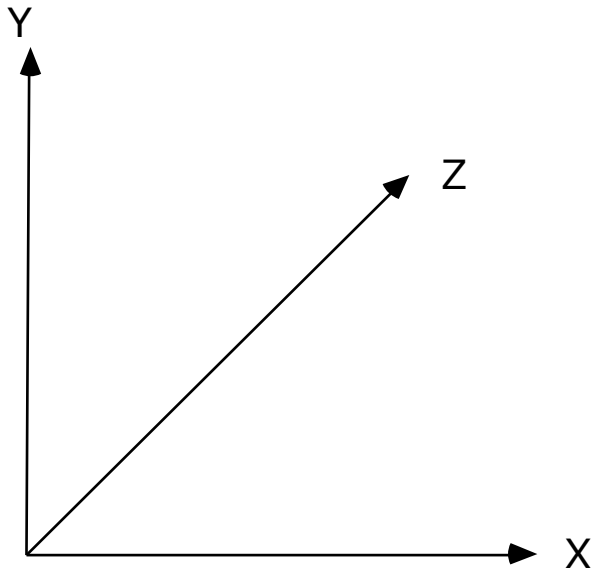
Mathematics in Computer Graphics

Contents

- Geometry (2D, 3D)
- Trigonometry
- Vector spaces
 - Points, vectors, and coordinates
- Dot and cross products
- Linear transforms and matrices

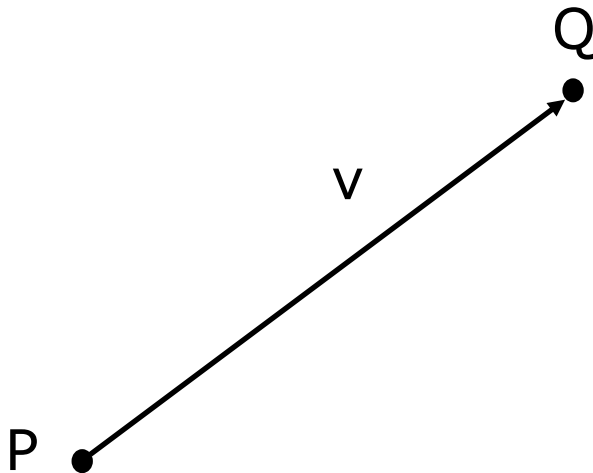
Coordinate Systems

- Orthogonal coordinate systems
 - Left-hand
 - Right-hand



Points

- 2D: $P(x, y)$, 3D: $P(x, y, z)$
- Point-point subtraction: $Q - P = v$
 - Result is a vector pointing from P to Q
- Vector-point addition: $P + v = Q$
 - Result is a new point



Line

□ Slope based equation

$$y = mx + b$$

$$\text{Slope } m = (y - y_1) / (x - x_1) = (y_2 - y_1) / (x_2 - x_1)$$

□ Parametric equation

$$P = P_1 + t(P_2 - P_1), \text{ as vector based definition}$$

or

$$P = (1-t)P_1 + t P_2, \text{ as affine combination of two points}$$

$$x = x_1 + t(x_2 - x_1)$$

$$y = y_1 + t(y_2 - y_1)$$

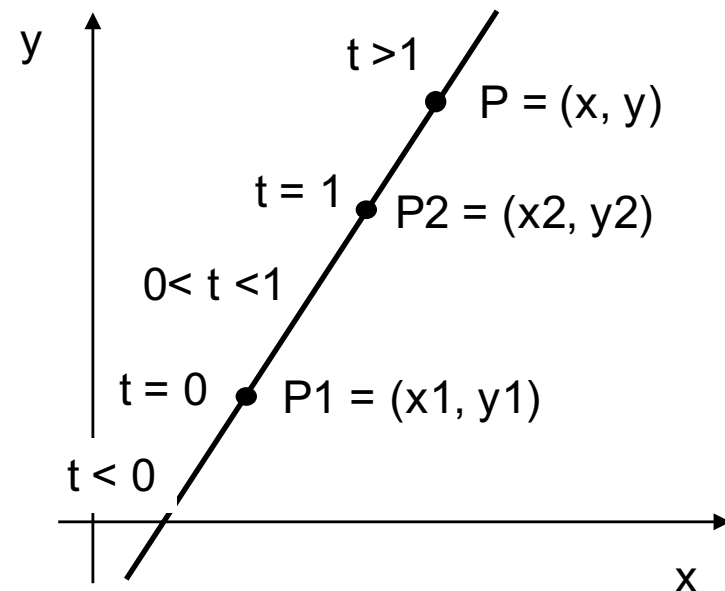
$t < 0$, before $P_1(x_1, y_1)$

$t = 0$, $P = P_1$

$0 < t < 1$, get points
between P_1 and P_2

$t = 1$, $P = P_2(x_2, y_2)$

$t > 1$, after P_2



Line related formulas

- Length = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Midpoint, p_3 , between p_1 and p_2
$$p_3 = ((x_1 + x_2) / 2, (y_1 + y_2) / 2)$$
- Two lines are perpendicular if:
$$m_1 = -1/m_2$$
$$\cos(\theta) = 0$$

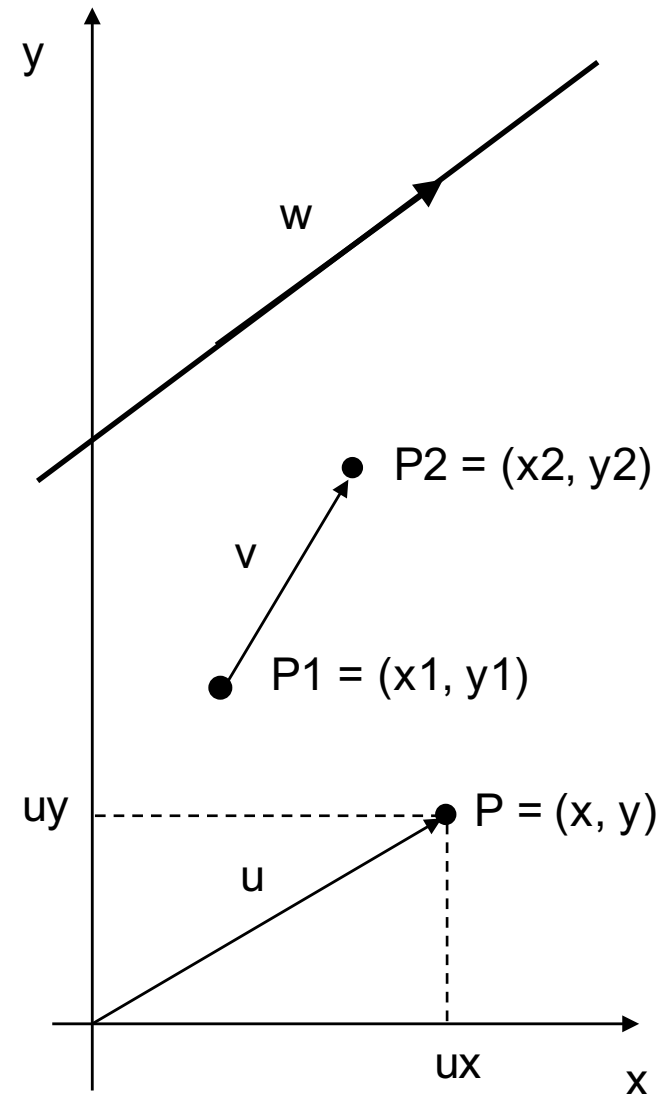
dot product of related vectors is zero ($u \bullet v = 0$)
- Point P_3 lays on the line P_1P_2 if:
Cartesian space, slope based definition:
$$y_3 = mx_3 + b, \text{ where } m(P_1, P_2), b(P_1, P_2)$$

Affine space, parametric definition:
$$P_3 = P_1 + t(P_2 - P_1), \text{ the system of equation on } x, y \text{ (and } z)$$

should have solution. The position depends on the t value
($<0, 0, 0..1, 1, 1<$)

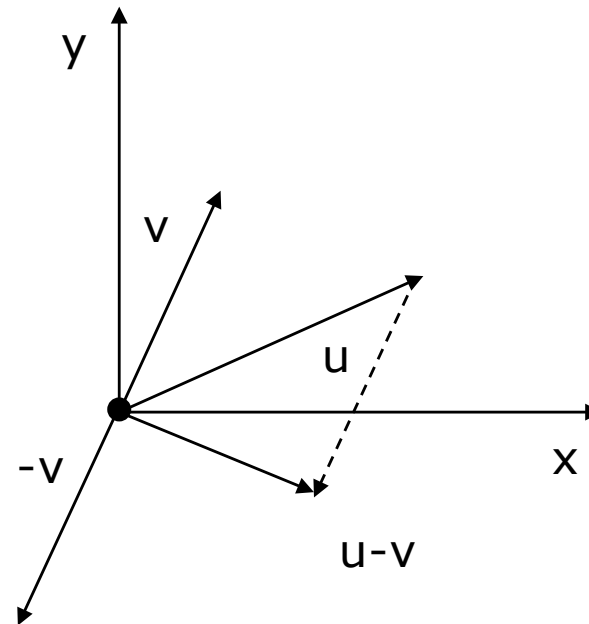
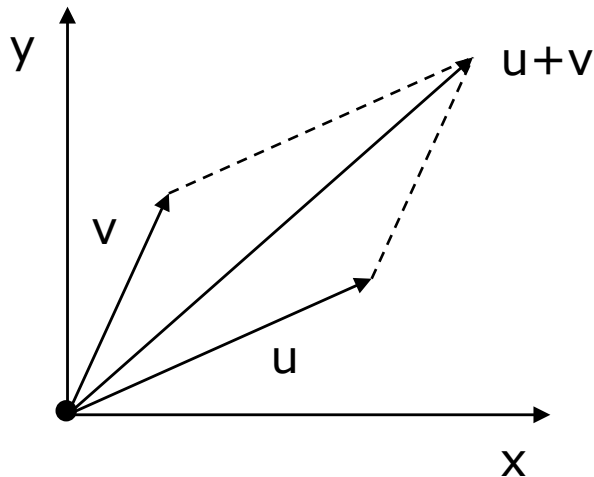
Vectors

- Two types of elements:
 - Scalars (real numbers): $a, b, g, d, \dots$
 - Vectors (n-tuples): $u, v, w, \dots$
- Used for:
 - Points in space (i.e., location)
 - Displacements from point to point
 - Direction (i.e., orientation)
- Operations:
 - Addition
 - Subtraction
 - Dot Product
 - Cross Product
 - Norm



Vector operations

- Addition
- Subtraction
- Multiplication by a scalar
 - Distributive rule: $a(u + v) = a(u) + a(v)$
 $(a + b)u = au + bu$
 - “stretches” a vector, changing its length but not its direction



Dot product

- The dot product or, more generally, inner product of two vectors is a scalar:

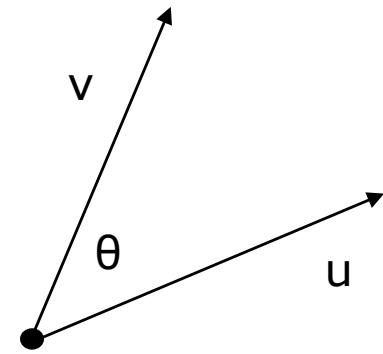
$$u \bullet v = u_x v_x + u_y v_y + u_z v_z \quad (\text{in 3D})$$

- Is commutative

$$u \bullet v = v \bullet u$$

- Is distributive with respect to addition

$$u \bullet (v + w) = u \bullet v + u \bullet w$$



- Used for:

- Computing the length (Euclidean Norm) of a vector:
 $\text{length}(v) = ||v|| = \text{sqrt}(v \bullet v)$
- Computing the distance between two points P and Q:
 $v = Q - P, \text{distance}(P, Q) = \text{length}(v) = ||v|| = \text{sqrt}(v \bullet v)$
- Normalizing a vector, making it unit-length: $v = v / ||v||$
- Checking two vectors for orthogonality
 $u \bullet v = 0$

Dot product

- Computing the angle between two vectors:

$$u \cdot v = |u| |v| \cos(\theta)$$

If $u \cdot v < 0$ then $\theta > 90^\circ$, $u \cdot v = 0$ then $\theta = 90^\circ$, $u \cdot v > 0$ then $\theta < 90^\circ$

- Relative position of a point P against a plane

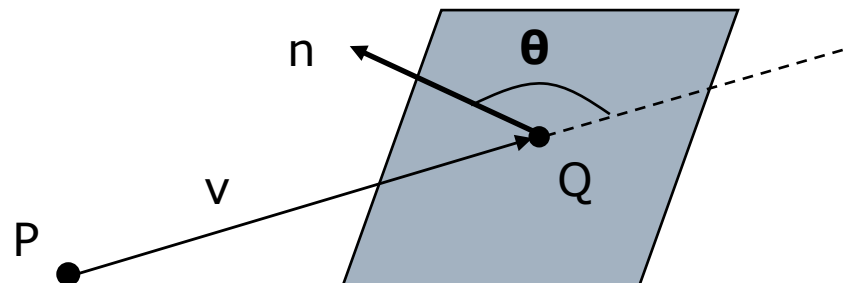
If the vector $v = Q - P$, where Q is any point on the plane, and the normal vector of the plane is n , v and n gives an angle:

$\theta > 90^\circ$ then P is in front of the plane

$\theta = 90^\circ$ then P is on the plane

$\theta < 90^\circ$ then P is on back of the plane

- If the viewer (P) is in front of the lateral plane polygon of a convex 3D object, the polygon is visible



Dot product

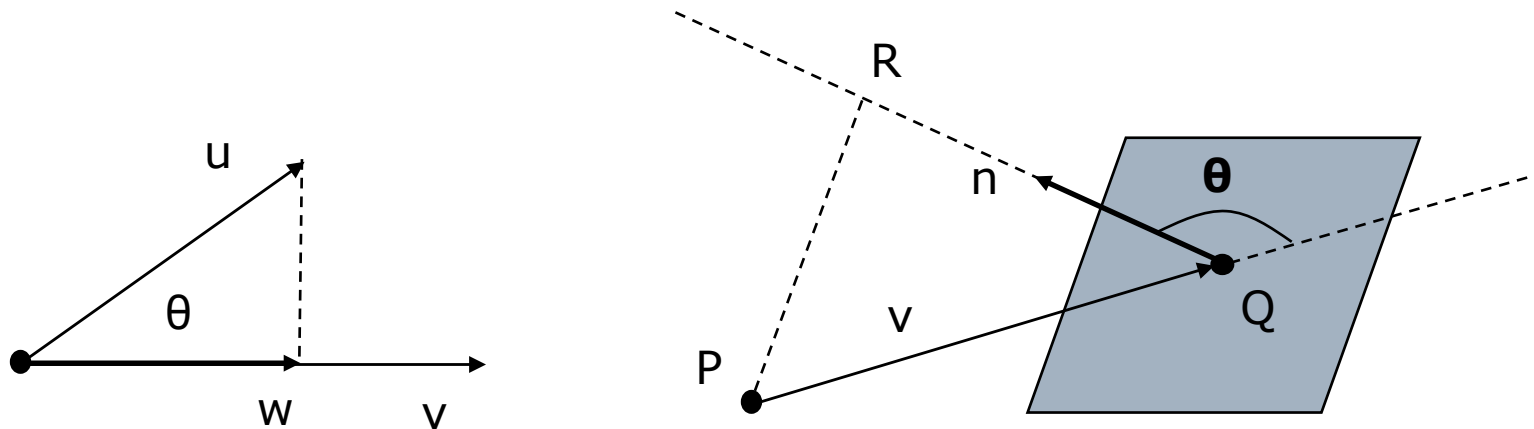
□ Projecting a vector onto another

- If v is a unit vector and we have another vector, u
- We can project u perpendicularly onto v
- And the result w has length $u \cdot v$

$$||w|| = ||u|| \cos(\theta) = ||u|| \left[\frac{u \cdot v}{||u|| \cdot ||v||} \right] = u \cdot v$$

□ Computes the distance between point P and a plane.

Normal vector is normalized ($||n||=1$), then distance = $||Q-R||$



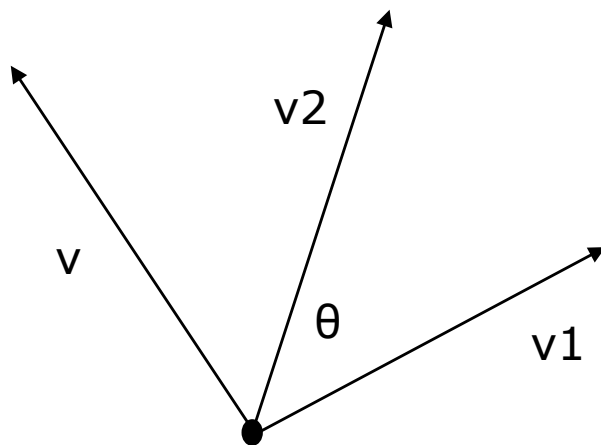
Cross product

- The cross product or vector product of two vectors is a vector orthogonal to both,

$$\mathbf{v}_1 \times \mathbf{v}_2 = n |\mathbf{v}_1| |\mathbf{v}_2| \sin(\theta),$$
 n is normal vector to both $\mathbf{v}_1$ and $\mathbf{v}_2$
- Right-hand rule dictates direction of cross product

$$\mathbf{v}_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \quad \mathbf{v} = \mathbf{v}_1 \times \mathbf{v}_2 = \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & 0 & 0 \\ 0 & y_2 & 0 \end{bmatrix}$$

$\mathbf{i}=(1,0,0), \mathbf{j}=(0,1,0), \mathbf{k}=(0,0,1)$ – cartesian system



$$\mathbf{v} = \mathbf{v}_1 \times \mathbf{v}_2 = \begin{bmatrix} y_1 z_2 - y_2 z_1 \\ -(x_1 z_2 - x_2 z_1) \\ x_1 y_2 - x_2 y_1 \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ x_1 y_2 \end{bmatrix}$$

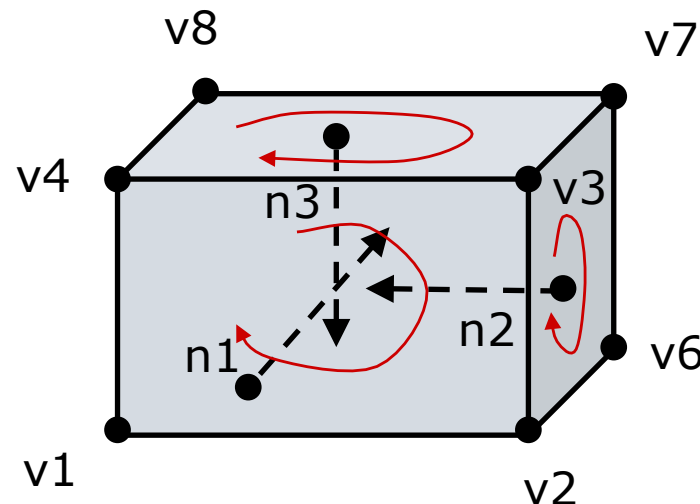
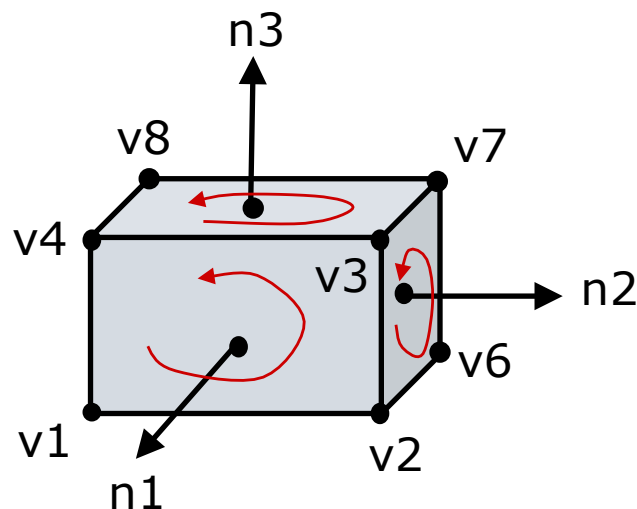
Cross product

□ Application:

cross product computes the normal vector to a plane polygon
 $n1 = e1 \times e2$, where $e1$ and $e2$ could be given by two consecutive edges of a polygon $v1v2$ and $v2v3$ respectively.

□ Convention

1. Normal vector points away of the object
 - Counterclockwise vertex sequence, e.g. $v1, v2, v3, v4$
2. Normal vector points into the object
 - Clockwise vertex sequence, e.g. $v1, v4, v3, v2$



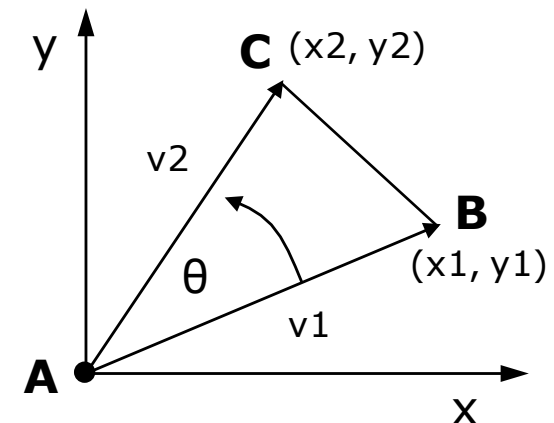
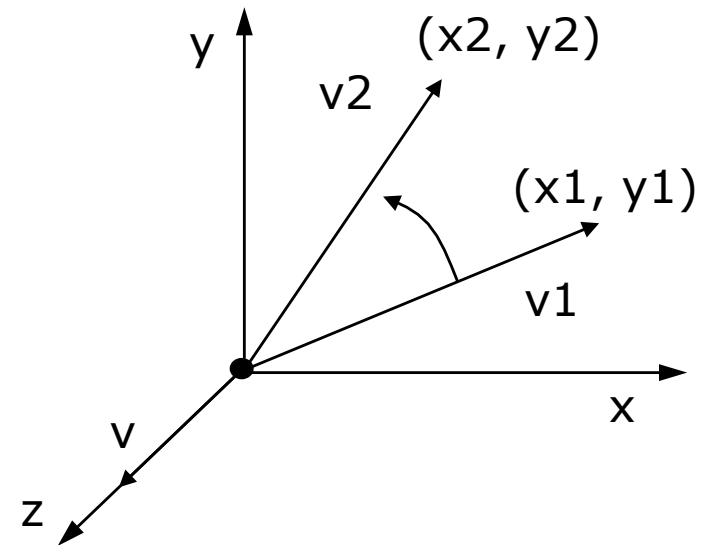
Triangle computation

- Consider the general case of the cross product:

$$v = v_1 \times v_2 = \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & y_1 & 0 \\ x_2 & y_2 & 0 \end{bmatrix}$$

$$v = (x_1 y_2 - x_2 y_1) \mathbf{k}$$

The cross product is a vector v along z ,
of module $|x_1 y_2 - x_2 y_1|$.



Area of a triangle

The triangle ABC has the area:

$$\begin{aligned} & \frac{1}{2} (x_1 - x_2) (y_1 + y_2) + \\ & \frac{1}{2} (x_2 - x_0) (y_2 + y_0) + \\ & \frac{1}{2} (x_0 - x_1) (y_0 + y_1) = \end{aligned}$$

$$\begin{aligned} & \frac{1}{2} (x_1 y_1 + x_1 y_2 - x_2 y_1 - x_2 y_2 + \\ & \quad x_2 y_2 + x_2 y_0 - x_0 y_2 - x_0 y_0 + \\ & \quad x_0 y_0 + x_0 y_1 - x_1 y_0 - x_1 y_1) = \end{aligned}$$

$$\frac{1}{2} (x_0 y_1 - x_1 y_0 + x_1 y_2 - x_2 y_1 + x_2 y_0 - x_0 y_2)$$

where $x_0, y_0 = 0$

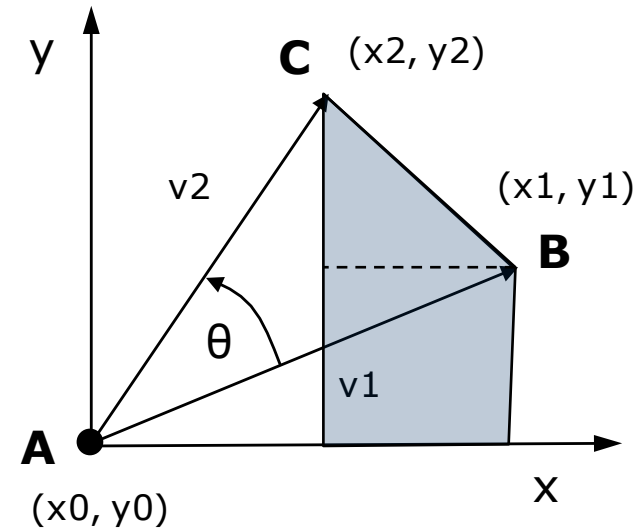
$$S = \frac{1}{2} (x_1 y_2 - x_2 y_1)$$

therefore

$$S = \frac{1}{2} |v_1 \times v_2|$$

Moreover

$$v_1 \times v_2 = |v_1| |v_2| \sin \theta$$



Triangle computation

- Consider a triangle, (a, b, c)

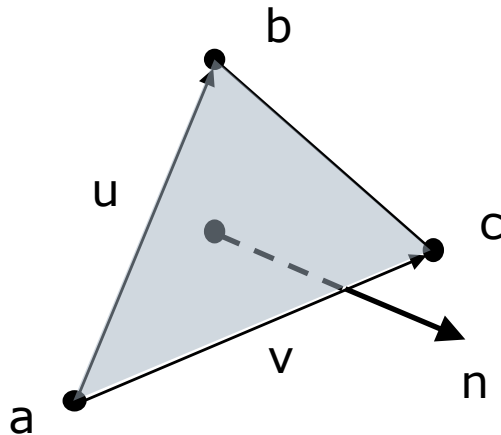
$a, b, c = (x, y, z)$ tuples

- Surface area by cross product

$$S = \frac{1}{2} ||u \times v|| = \frac{1}{2} * ||(b-a) \times (c-a)||$$

- Unit normal vector

$$n = (1/2S) * (b-a) \times (c-a)$$



Area of a polygon

Polygonal area:

$$\frac{1}{2} (x_0 - x_1) (y_0 + y_1) +$$

$$\frac{1}{2} (x_1 - x_2) (y_1 + y_2) +$$

...

$$\frac{1}{2} (x_n - x_0) (y_n + y_0) =$$

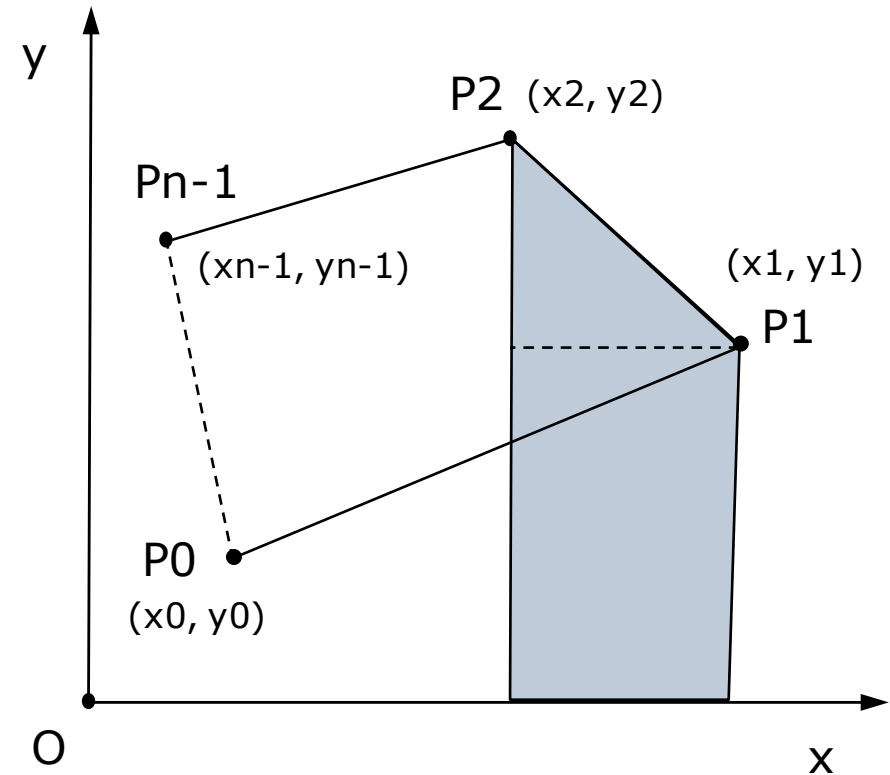
$$\frac{1}{2} (x_0 y_0 + x_0 y_1 - x_1 y_0 - x_1 y_1 +$$

$$x_1 y_1 + x_1 y_2 - x_2 y_1 - x_2 y_2 +$$

...

$$x_n y_n + x_n y_0 - x_0 y_n - x_0 y_0)$$

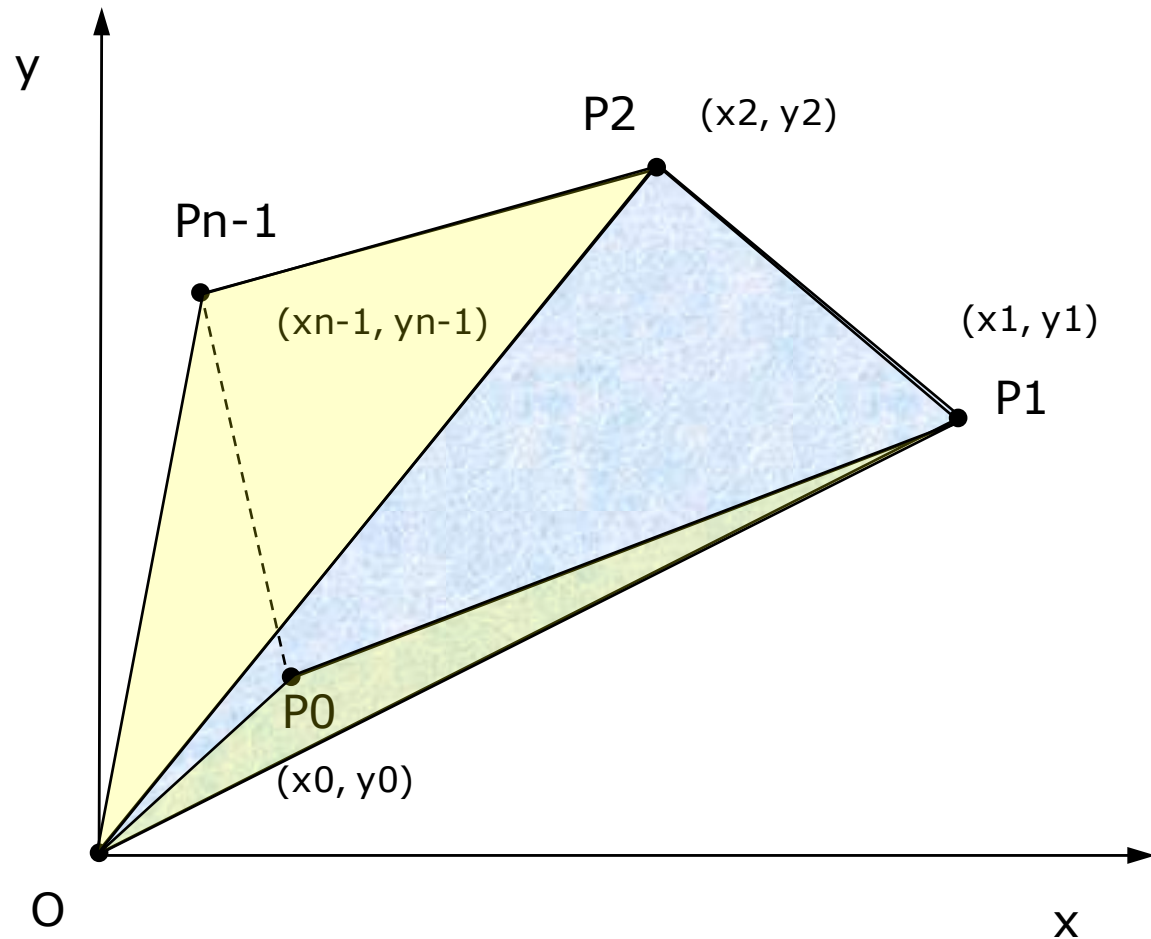
$$S = \frac{1}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i)$$



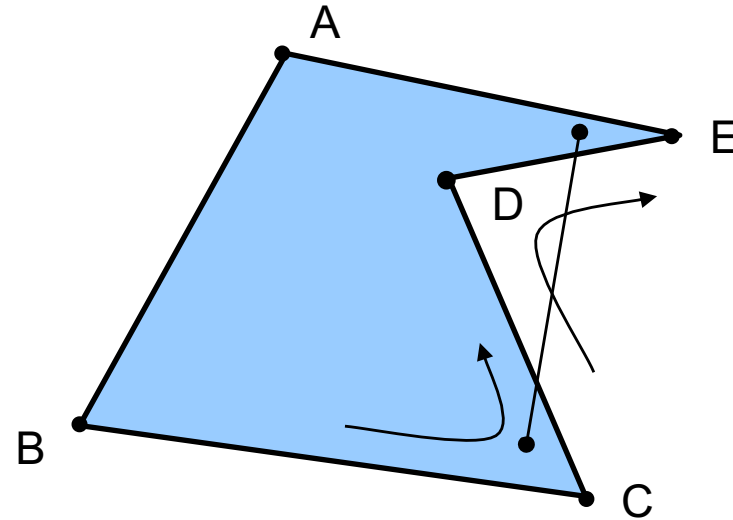
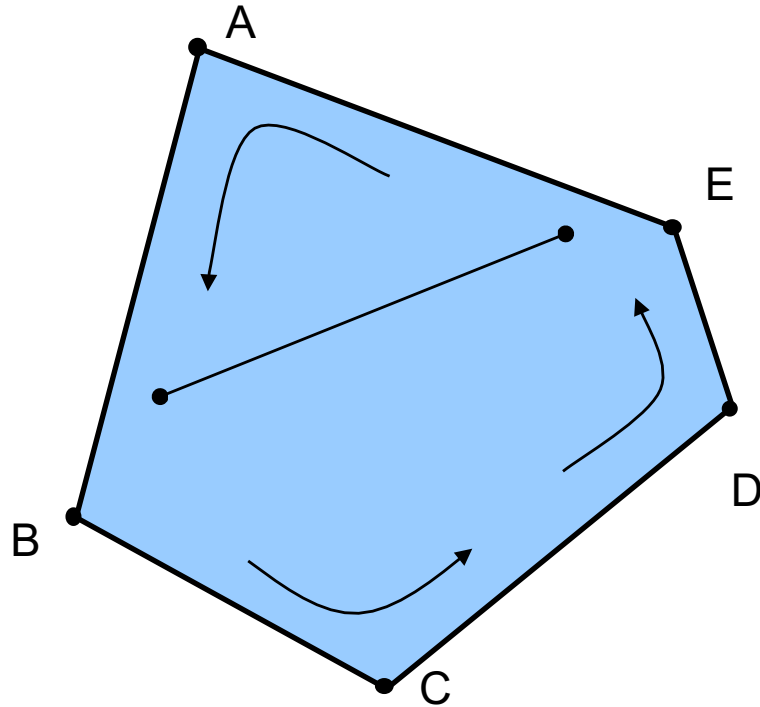
Area of a polygon

Polygonal area:

$$S = 1/2 \sum_{i=0}^{n-1} (v_i \times v_{i+1})$$



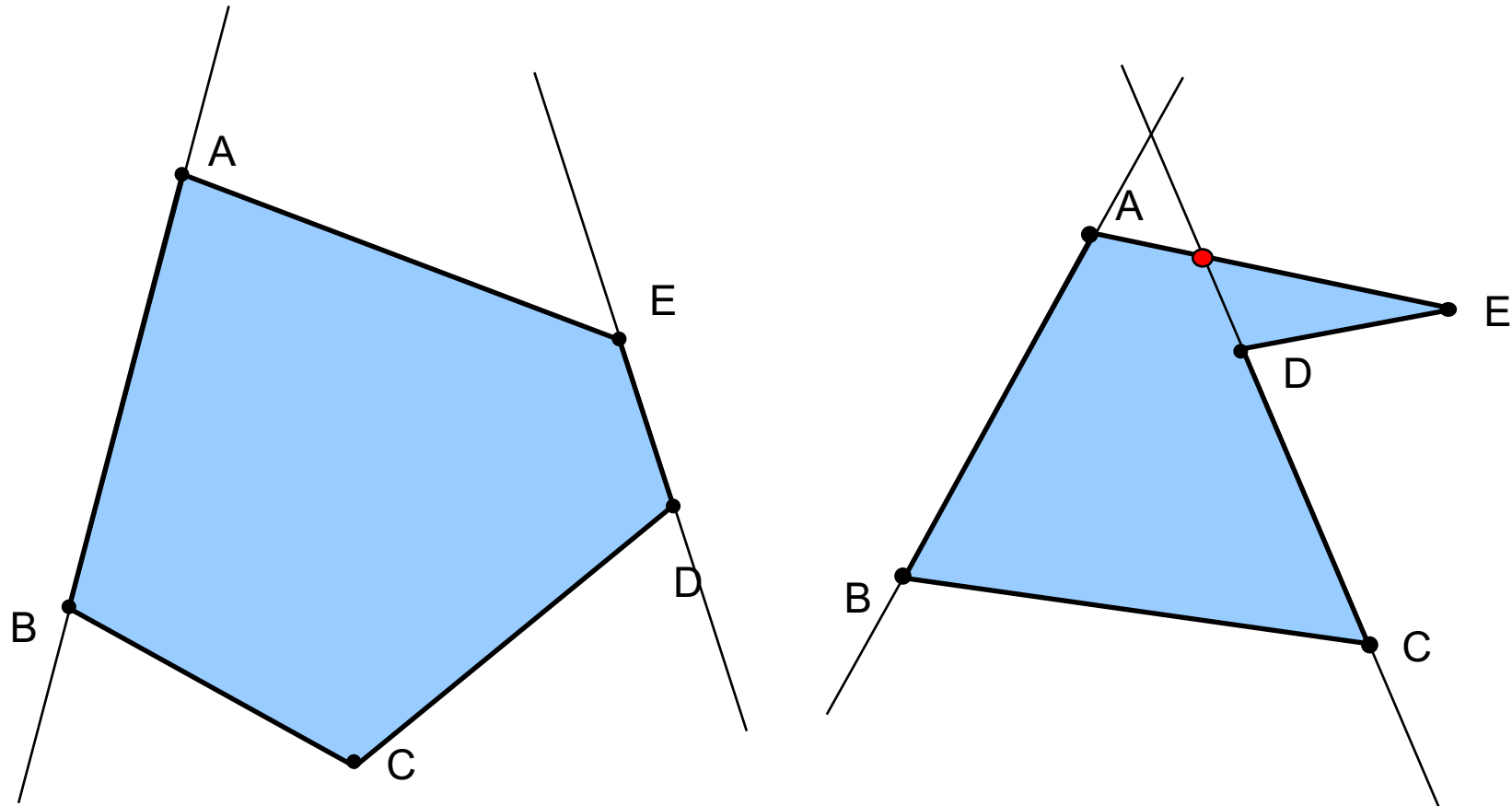
Convex and concave polygon



- ❑ Cross product of successive edges ($n = e_1 \times e_2 \sin(\theta)$)
- ❑ Turn directions
- ❑ Order of three successive vertices
- ❑ Angles
- ❑ Intersection between an edge and the polygon

Convex and concave polygon

- At least one intersection between an edge and the polygon



Test for convex and concave polygons

For the convex polygon all the cross products of adjacent edges will be the same sign.

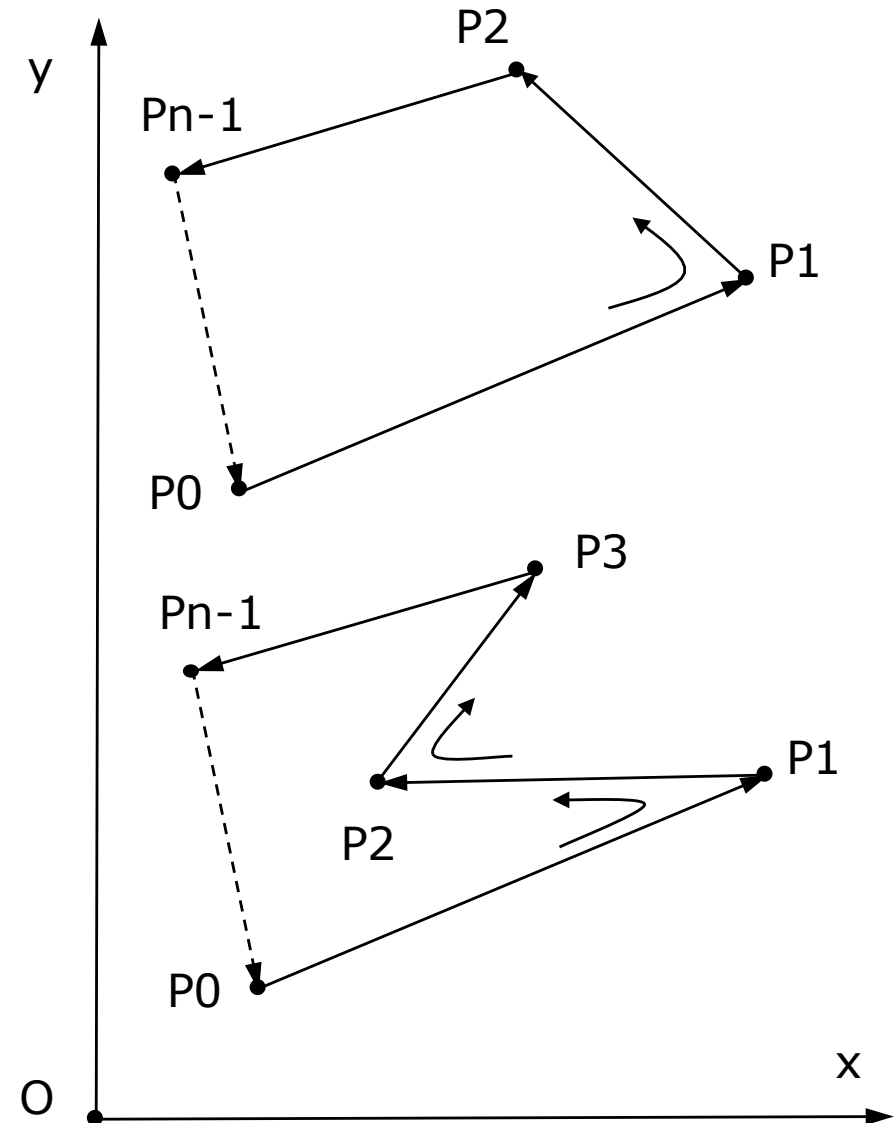
$$\begin{aligned}\text{sign}(e_1 \times e_2) &= \text{sign}(e_2 \times e_3) = \dots \\ &= \text{sign}(e_n \times e_1)\end{aligned}$$

Obs: moving along the edges of the convex polygon it always turns left.

The concave polygon has a mixture of cross product signs.

$$\begin{aligned}\text{NO: } \text{sign}(e_1 \times e_2) &= \text{sign}(e_2 \times e_3) = \dots \\ &= \text{sign}(e_n \times e_1)\end{aligned}$$

Obs: moving along the edges of the concave polygon it sometimes turns right.



Plane in Cartesian Space

- Plane equation:

$$Ax + By + Cz + D = 0$$

- Normal vector:

$$N [A, B, C]$$

- Equation by one point $P(x_0, y_0, z_0)$ and normal vector N :

$$A(x-x_0) + B(y-y_0) + C(z-z_0) + D = 0$$

- Plane defined by 3 points:

$$P(x_1, y_1, z_1), Q(x_2, y_2, z_2), R(x_3, y_3, z_3)$$

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$
$$A = \begin{vmatrix} y_1 & z_1 & 1 \\ y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \end{vmatrix}$$
$$B = - \begin{vmatrix} x_1 & z_1 & 1 \\ x_2 & z_2 & 1 \\ x_3 & z_3 & 1 \end{vmatrix}$$
$$C = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$
$$D = - \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

Plane in Affine Space

- Parametric plane equation by 3 points P, Q, and R:

$$(1-s)((1-t)P + tQ) + sR$$

$$\text{Or another form } P + t(Q-P) + s(R-(P + t(Q-P)))$$

- Equation of the plane through P and normal vector v as dot product:

$$(X - P) \bullet v = 0$$

- Intersection of a line with the plane:

$$\text{Plane equation } (X - P) \bullet v = 0$$

$$\text{Line equation } Q(t) = Q + tw$$

Replace the point X by Q(t) for the intersection

$$(Q + tw - P) \bullet v = 0$$

$$\text{Gives } t = \frac{((P - Q) \bullet v)}{w \bullet v}$$

$$\text{And the intersection point is } Q + \frac{((P - Q) \bullet v)}{w \bullet v} w$$

Distance to a line

- Line equation is:

$P(t) = P_0 + t v$, t parameter, v direction vector

- Distance from R to the line:

It is the point $P(t)$ for which the distance is minimum

$(R - P(t)) \cdot v = 0$, and expanding

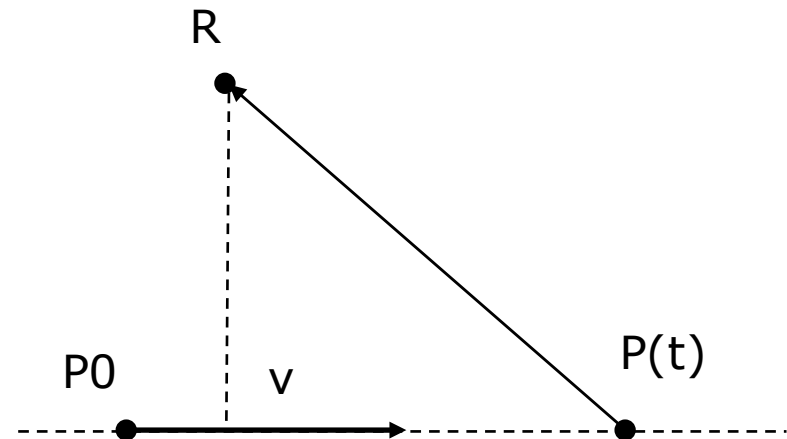
$(R - P_0 - t v) \cdot v = 0$

$(R - P_0) \cdot v = t v \cdot v$

Result
$$t = \frac{(R - P_0) \cdot v}{v \cdot v}$$

And the closest point to R is:

$$P_0 + \frac{(R - P_0) \cdot v}{v \cdot v} v$$



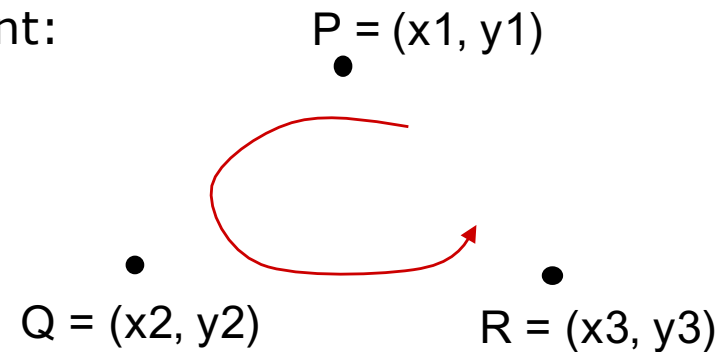
Point to point relationship

□ Ordered points

P, Q, and R counterclockwise order

The order expressed as a determinant:

$$D = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$



If $\text{sign}(D)=1 (+)$ then counterclockwise direction

$\text{sign}(D)=0$, collinear points

$\text{sign}(D)=-1 (-)$, clockwise direction

Point to line relationship

□ $P(x_0, y_0)$ on the line

P checks the equation: $P = P_1 + t(P_2 - P_1)$

The system should have a real solution on t :

$$x_0 = x_1 + t(x_2 - x_1)$$

$$y_0 = y_1 + t(y_2 - y_1)$$

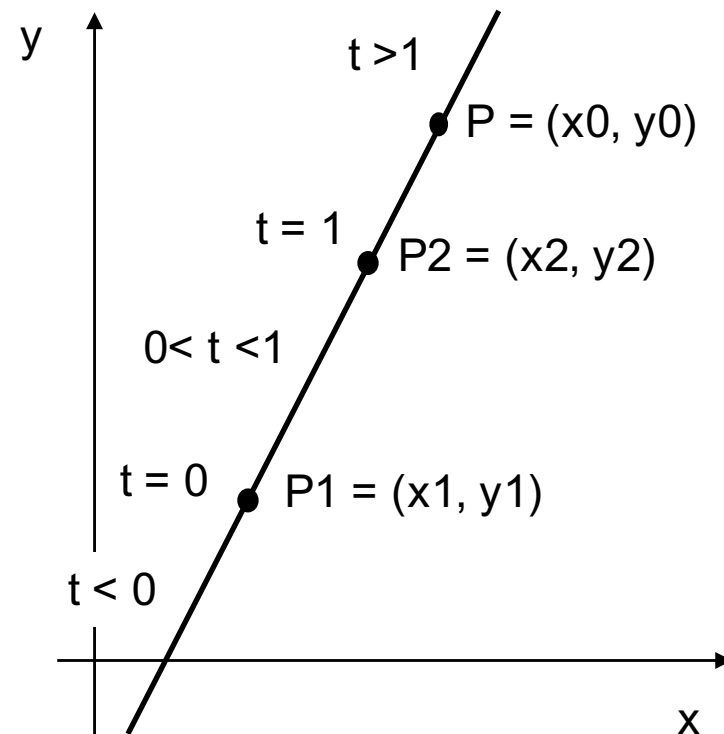
Otherwise P is outside of the line

Discussion on the t :

$$t=0, \quad P=P_1(x_1, y_1)$$

$$t=1, \quad P=P_2(x_2, y_2)$$

$0 < t < 1$, the P position
between P_1 and P_2



Point to triangle relationship

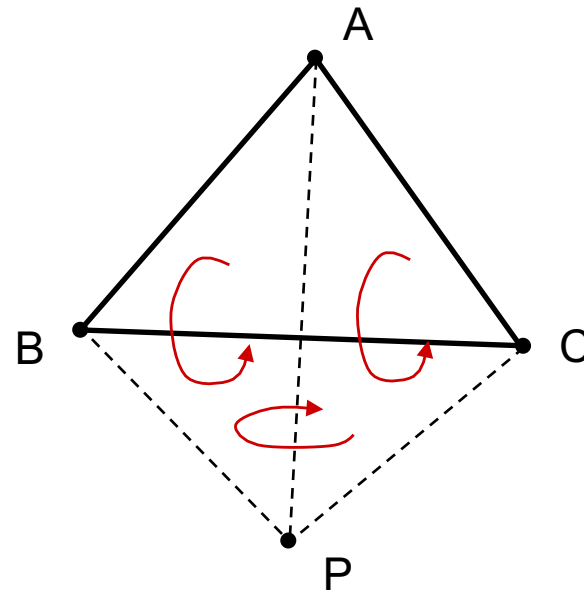
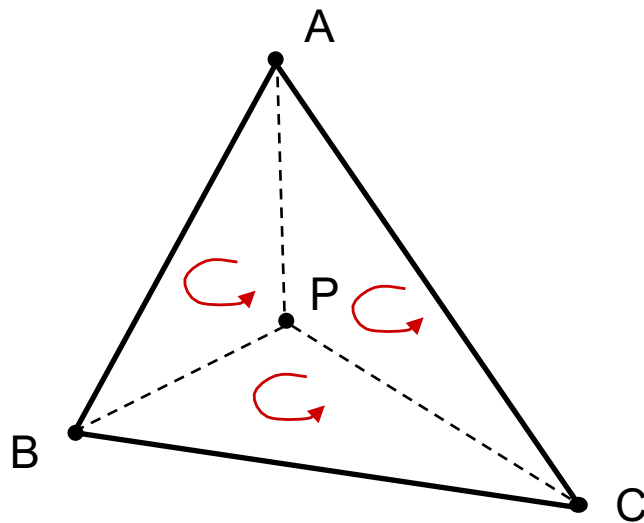
- Compute the order relationships of P against the triangle vertices A, B, and C

- Inside of the triangle

$$\text{sign}(A,B,P)=\text{sign}(B,C,P)=\text{sign}(C,A,P)$$

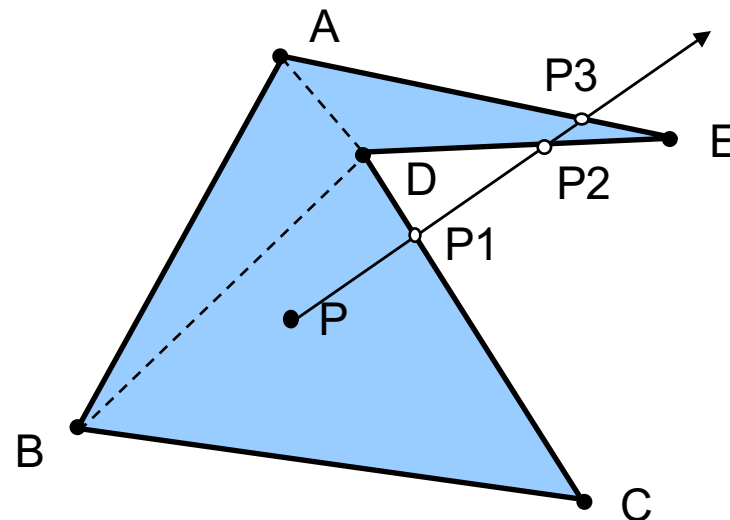
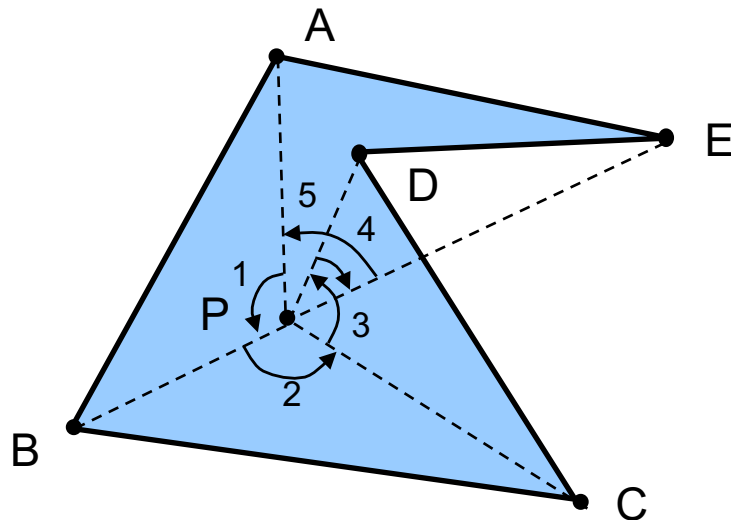
- Outside of the triangle

$$\text{NO: } \text{sign}(A,B,P)=\text{sign}(B,C,P)=\text{sign}(C,A,P)$$

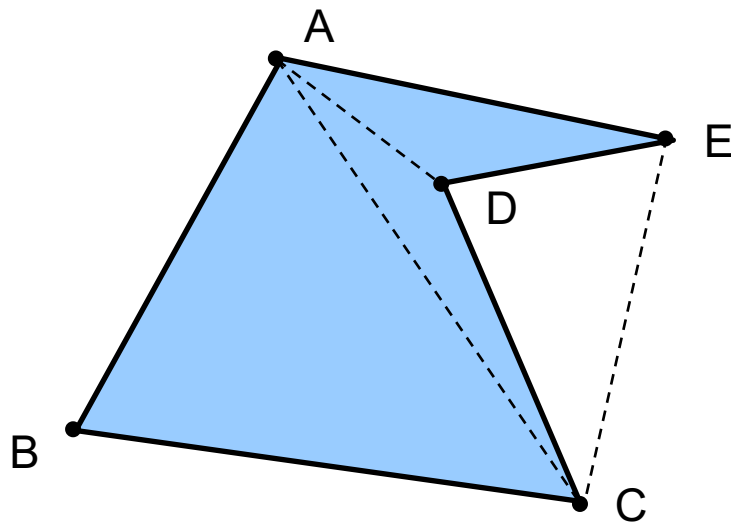


Point to polygon relationship

- P lays inside the polygon
 - Sum of angles given by the edges is 360°
 $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 = 360^\circ$
 - P lays inside at least one triangle of the triangulation set
 $\triangle ABD, \triangle BDC, \triangle DEA, P \in \triangle ABD$
 - Number of intersection between a line starting on P and the triangle
Odd number of intersection: P1, P2, P3

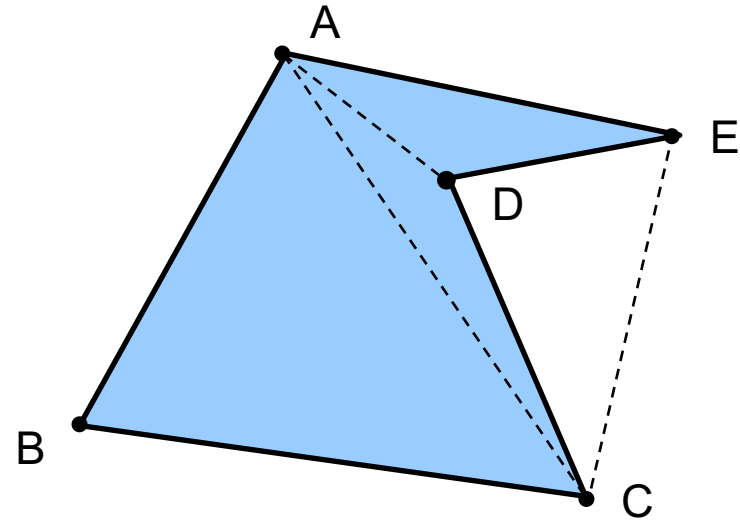
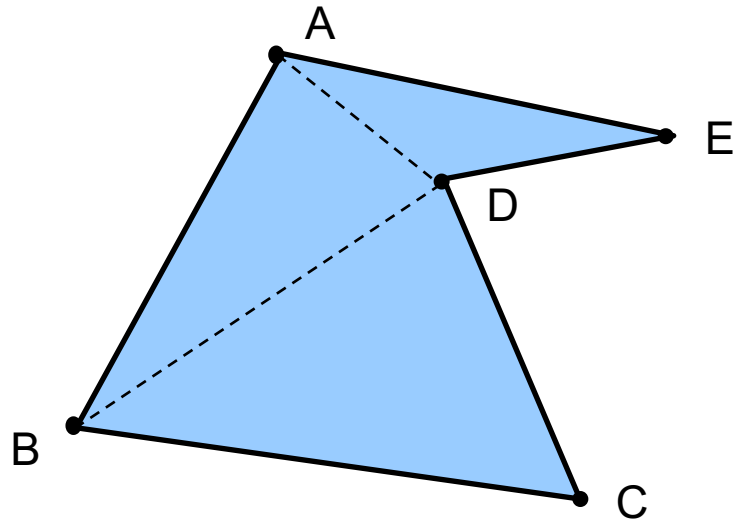


Polygon triangulation



1. Input circular list: $IL = A, B, C, D, E$
Output list of triangles: OL
2. $IL = A, B, C, D, E, OL = \emptyset$,
Analyze $\triangle ABC$, $\triangle ABC \blacksquare P$
 $OL \leftarrow \triangle ABC$, cut off B from IL
3. $IL = A, C, D, E, OL = \triangle ABC$,
analyze $\triangle CDE$, $\triangle CDE \blacksquare P$
 $OL \leftarrow \emptyset$
4. $IL = A, C, D, E, OL = \triangle ABC$,
Analyze $\triangle ACD$, $\triangle ACD \blacksquare P$
 $OL \leftarrow \triangle ACD$, cut off C from IL
5. $IL = A, D, E, OL = \triangle ABC, \triangle ACD$
Analyze $\triangle ADE$, $\triangle ADE \blacksquare P$
 $OL \leftarrow \triangle ADE$, cut off D from IL
6. $IL = A, E$, then stop
 $OL = \triangle ABC, \triangle ACD, \triangle ADE$

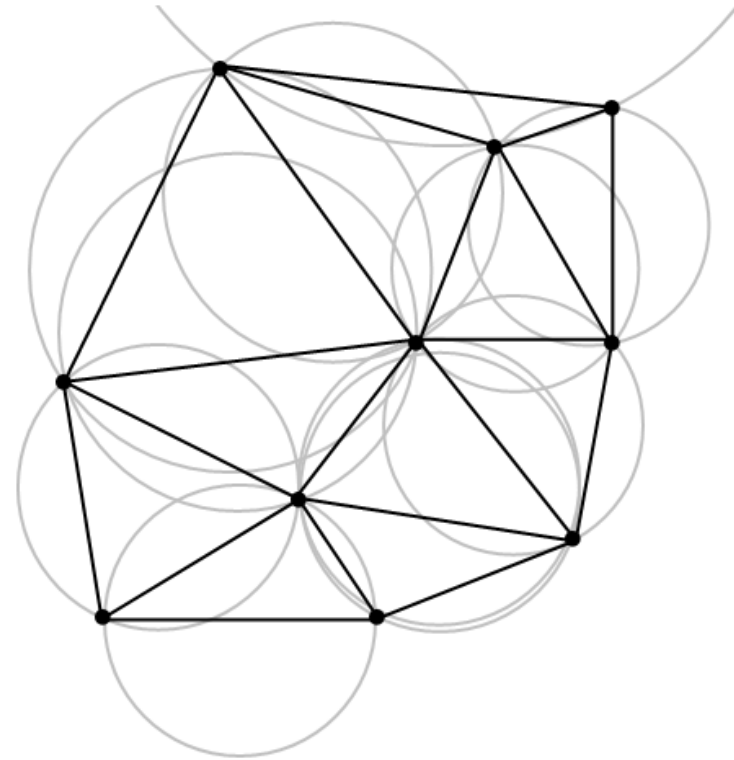
Polygon triangulation



- Another approach: Delaunay triangulations

Delaunay triangulation

- Delaunay triangulation:
for a set P of points in the plane is a triangulation $DT(P)$ such that no point in P is inside the circumcircle of any triangle in $DT(P)$.
- Delaunay triangulations maximize the minimum angle of all the angles of the triangles in the triangulation. They tend to avoid skinny triangles.
- The triangulation was invented by Boris Delaunay in 1934.



Point to plane relationship

- A very general test to determine if a point P is “inside” a plane Γ , defined by the point Q and the normal vector n :

$$(P - Q) \cdot n < 0:$$

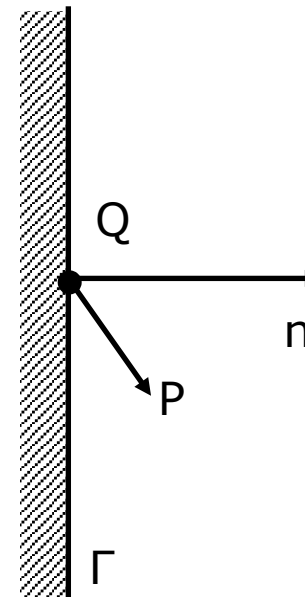
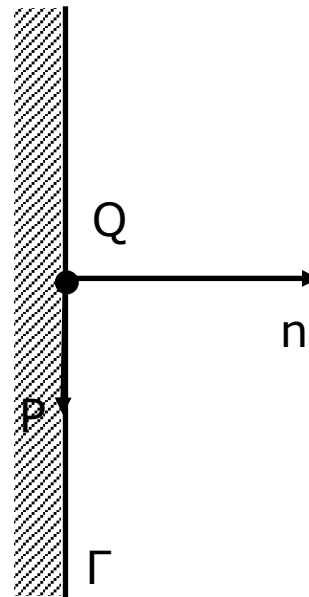
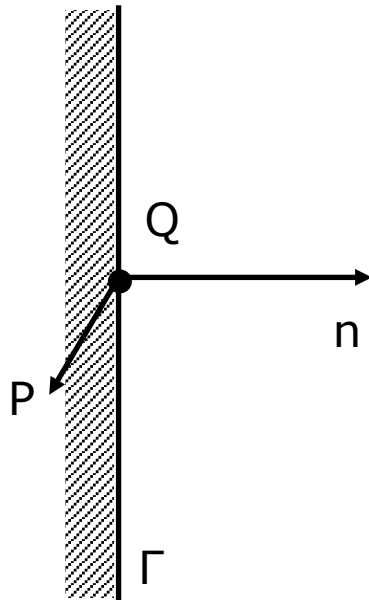
P inside Γ

$$(P - Q) \cdot n = 0:$$

p on Γ

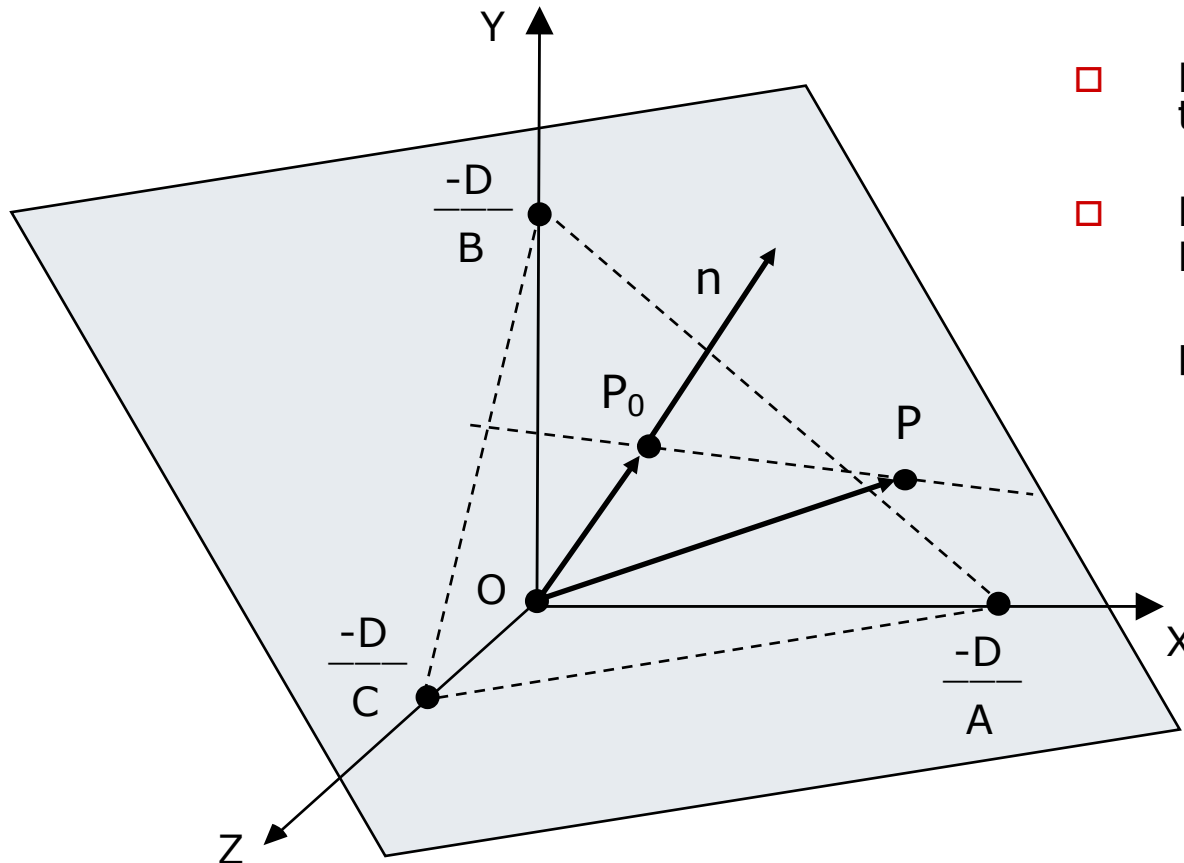
$$(P - Q) \cdot n > 0:$$

p outside Γ



Point to plane relationship

- Plane Γ : $Ax + By + Cz + D = 0$
- Normal vector n $[A \ B \ C]$
- Plane Γ intersects the axes on $-D/A$, $-D/B$ and $-D/C$
If the intersection points are on the positive axes then A , B , C are positive and D is negative



- Distance from the origin O to the plane Γ is $|D|$

- For function $F(x,y,z) = Ax + By + Cz + D$,

position of the point $Q(x_q, y_q, z_q)$ is:

if $F(Q) > 0$ then Q in front of the plane Γ (semispace of n)

if $F(Q) = 0$ then Q on the plane Γ

if $F(Q) < 0$ then Q behind the plane Γ

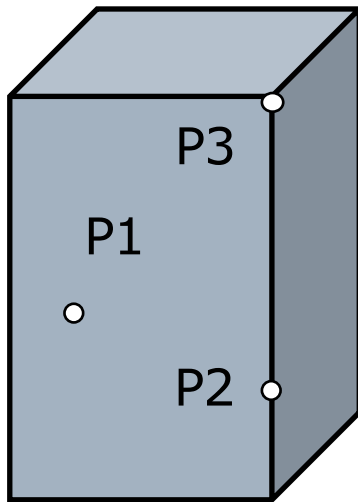
Point to 3D convex object

- Compute the plane equation and the function:

$$F_i(x,y,z) = A_i x + B_i y + C_i z + D_i$$

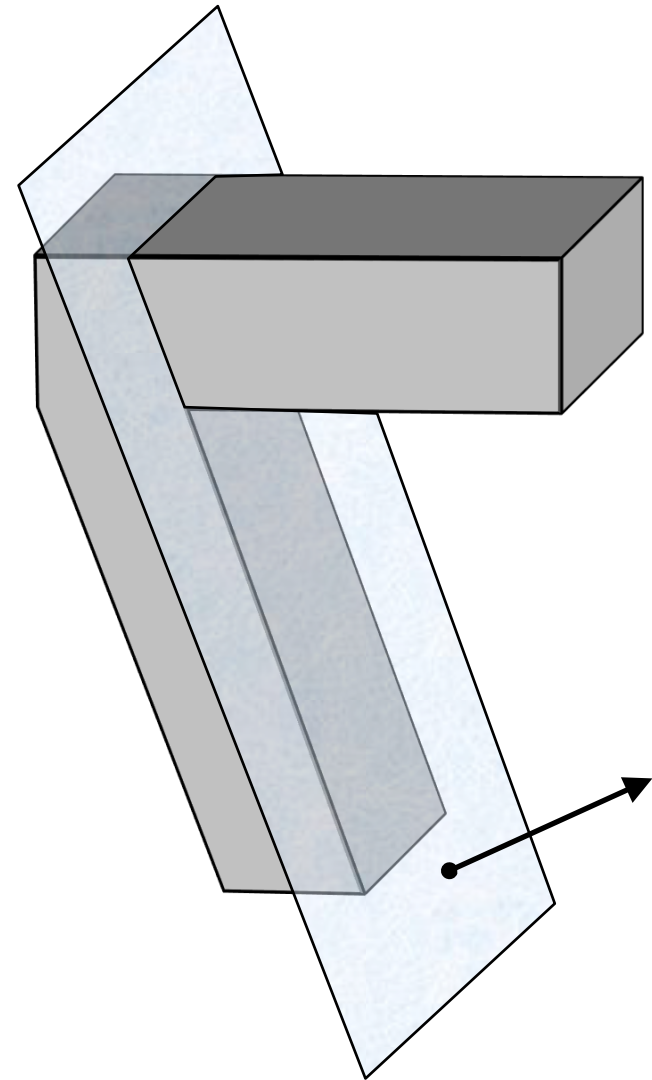
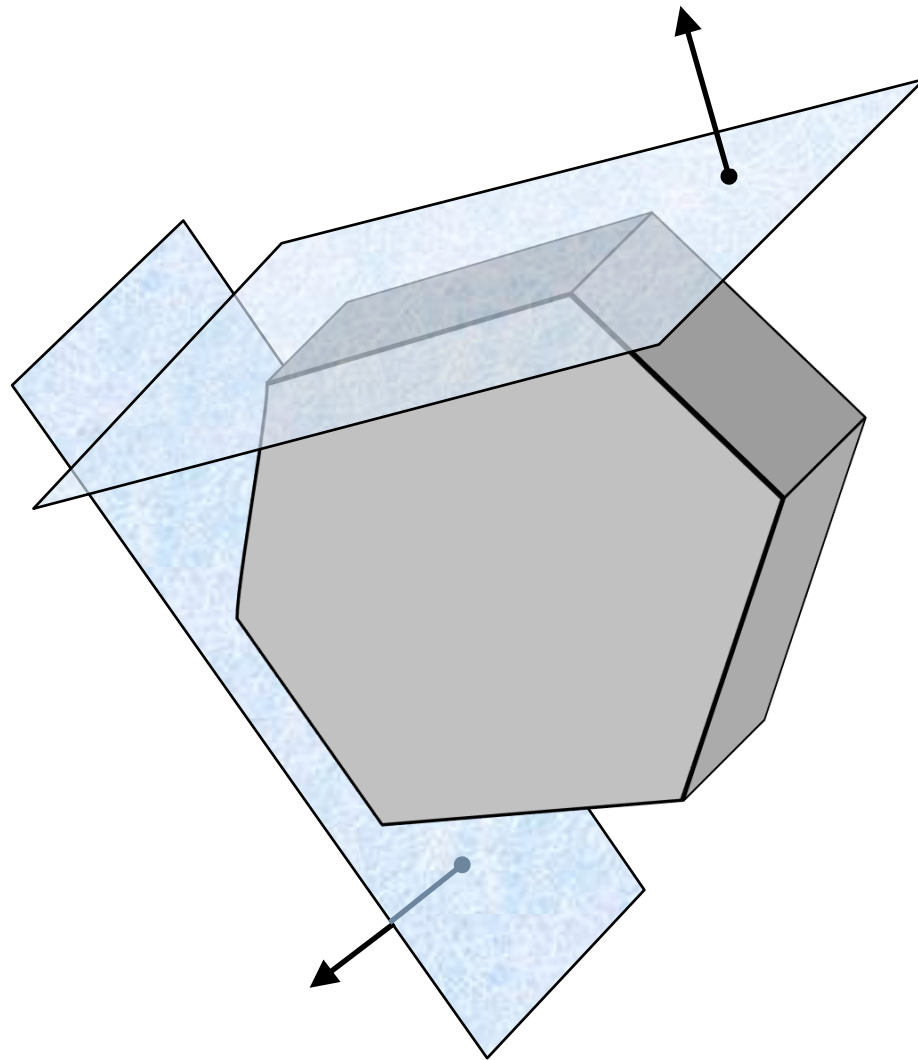
for all polygonal faces of the object Ω

- Compute the normal vectors to point away from the 3D polygonal object
- The point $P(x_p, y_p, z_p)$ is located as follows:



1. If at least one function gives $F(P) > 0$, then P outside of Ω
2. Else if one function gives $F_k(P) = 0$, then P on the k polygonal face
3. Else if two functions give $F_k(P) = F_q(P) = 0$, then P on the edge between the k and q polygonal faces
4. Else if r functions give $F_{k1}(P) = F_{k2}(P) = \dots = F_{kr}(P) = 0$, then P on the vertex between the $k1, k2, \dots, kr$ polygonal faces
5. Else if all functions give $F_k(P) < 0$, then P inside of Ω
6. Otherwise error

Convex and concave 3D object



Line to line relationship

- Resolve the system of two line equation

$$P = (1-t)P_1 + t P_2$$

$$P = (1-s)Q_1 + s Q_2$$

The equation should have real solution on t and s :

$$(1-t)P_1 + t P_2 = (1-s)Q_1 + s Q_2$$

In fact it is:

$$(1-t)x_{p1} + t x_{p2} = (1-s)x_{q1} + s x_{q2}$$

$$(1-t)y_{p1} + t y_{p2} = (1-s)y_{q1} + s y_{q2}$$

Discussion on the t and s :

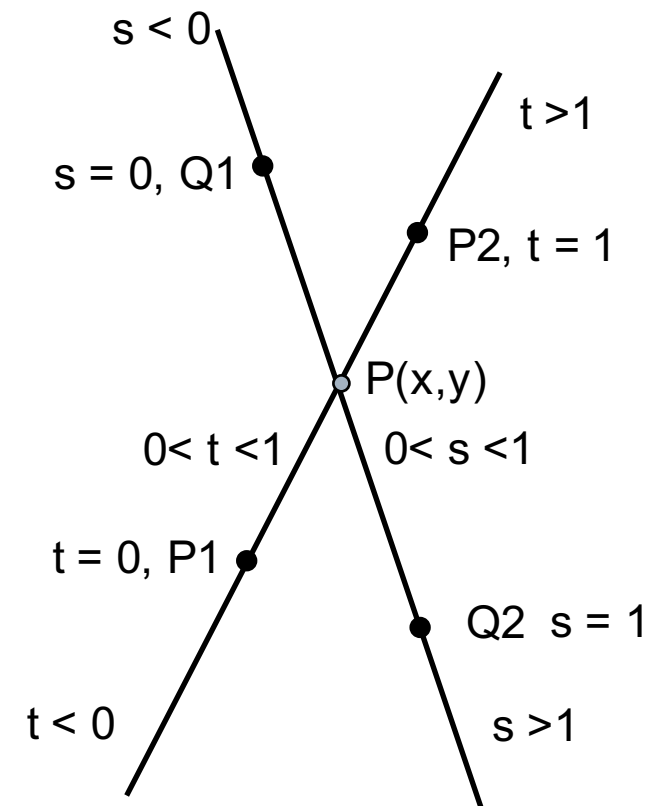
$t < 0, s < 0$ P before P_1 and Q_1

...

$0 < t < 1$ and $0 < s < 1$, P_1P_2 intersects Q_1Q_2

...

$t > 1, s > 1$ P after P_2 and Q_2



Line - plane intersection

- Parametric definition of edge:

$$L(t) = L_0 + (L_1 - L_0)t$$

If $t = 0$ then $L(t) = L_0$

If $t = 1$ then $L(t) = L_1$

Otherwise, $L(t)$ is part way from L_0 to L_1

- Edge intersects plane Γ where $E(t)$ is on Γ

Q is a point on Γ

n is the normal vector to Γ

$$(L(t) - Q) \cdot n = 0$$

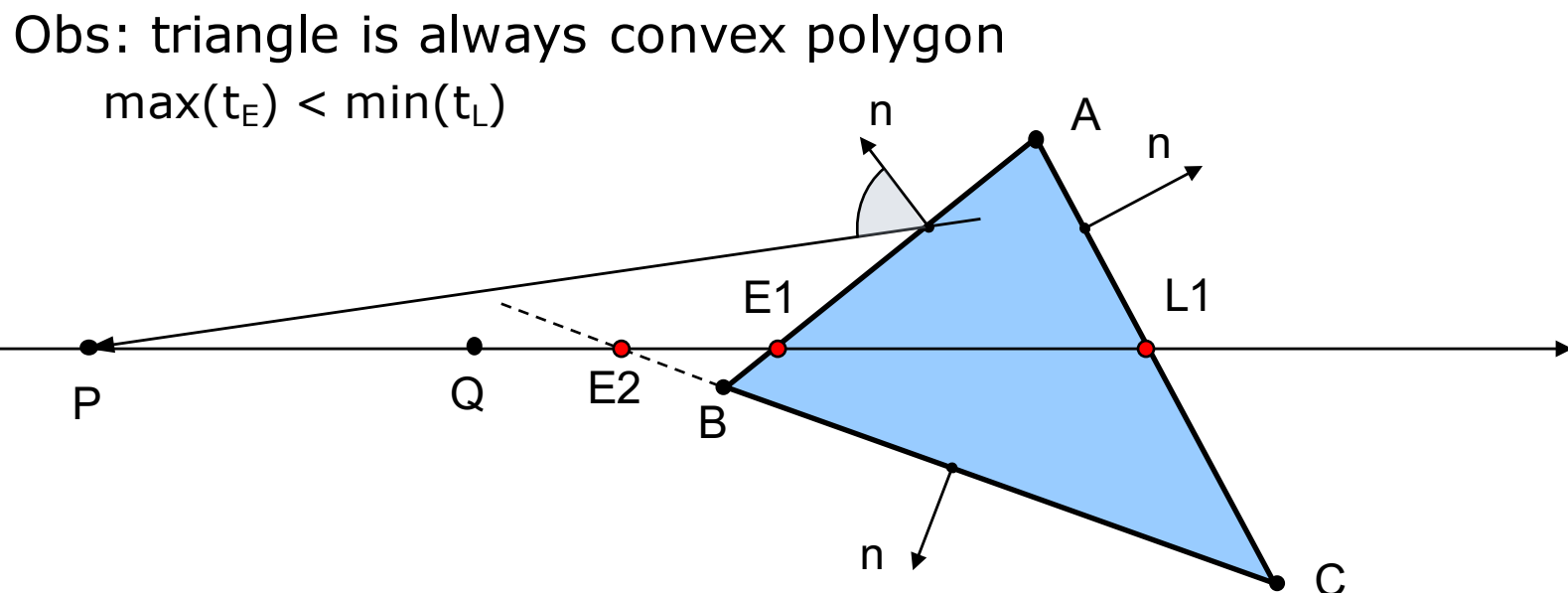
$$(L_0 + (L_1 - L_0)t - Q) \cdot n = 0$$

$$t = [(Q - L_0) \cdot n] / [(L_1 - L_0) \cdot n]$$

The intersection point $I = L(t)$ for this value of t

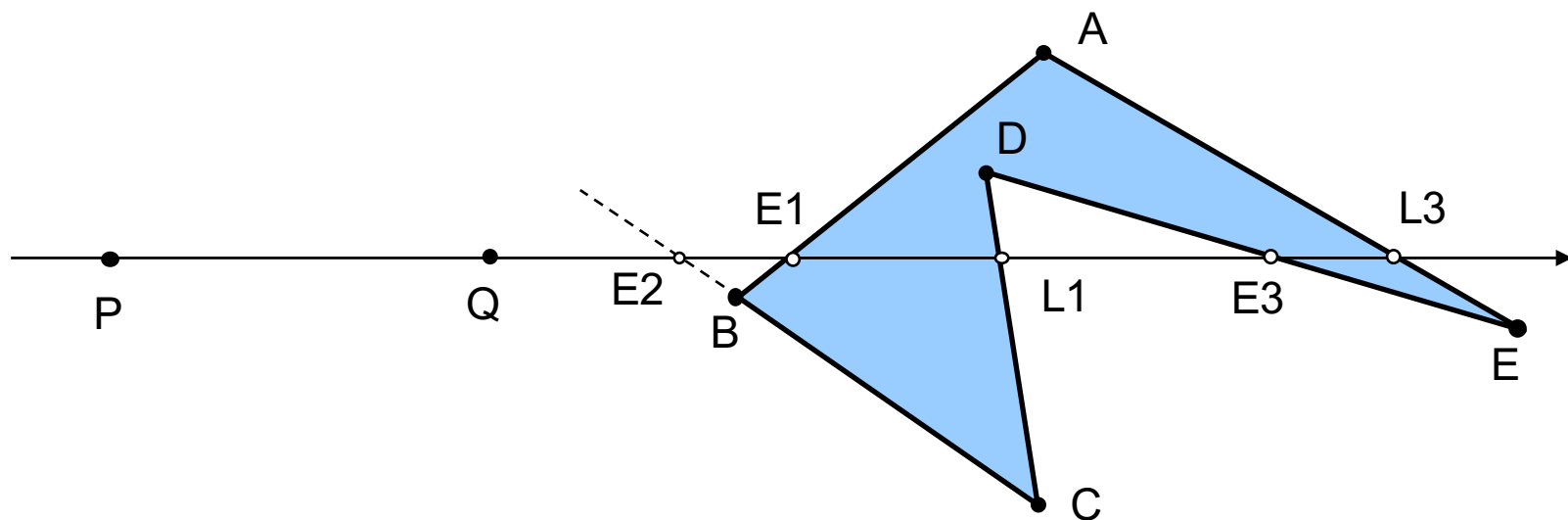
Line to triangle relationship

- P and Q inside the triangle
- P and Q outside
 - PQ intersects the triangle
 - PQ does not intersect the triangle
- Solutions:
 - Analyse the line to edges relationships
 - Cyrus-Beck algorithm

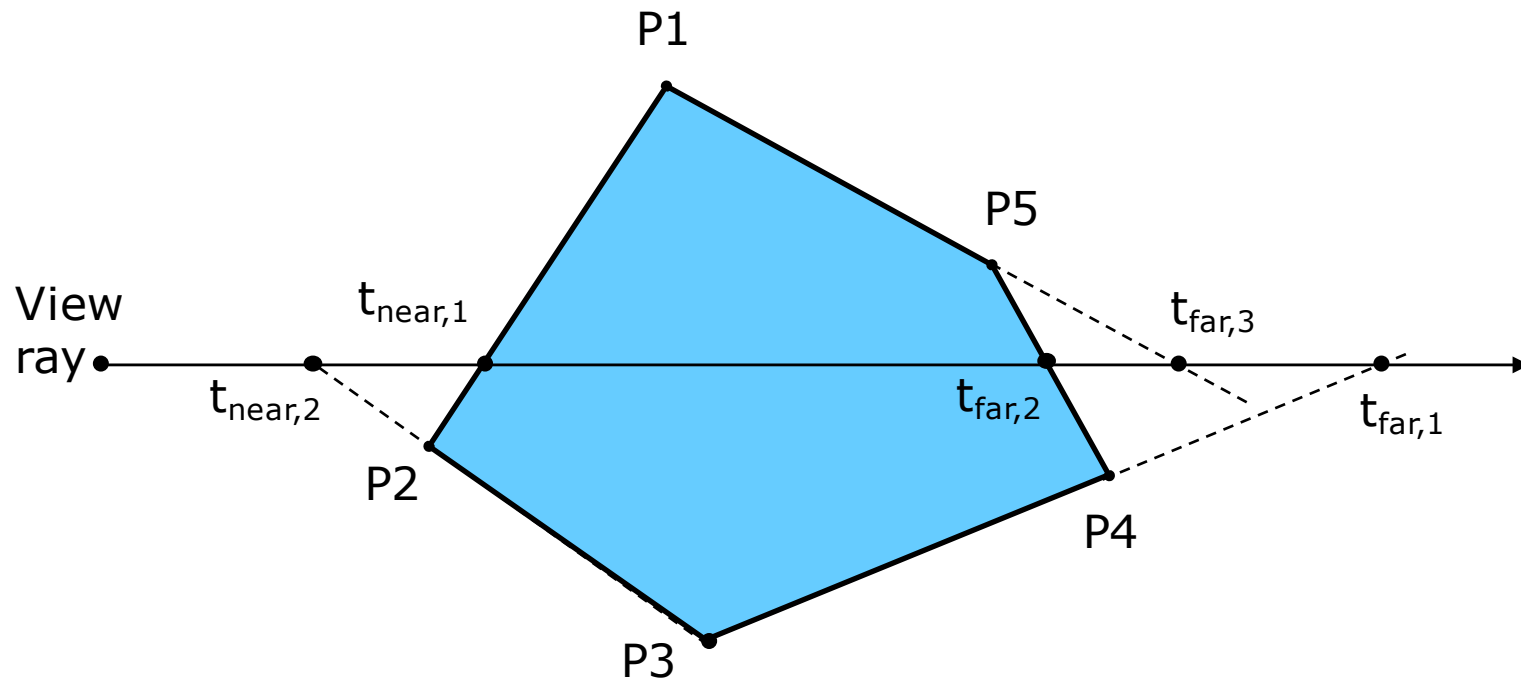


Line to polygon relationship

- ❑ Convex and concave polygons
- ❑ Triangulation
- ❑ Convex polygons
 - Analyses the line to edges relationships
 - Cyrus-Beck algorithm
(requires a convex polygon!)



Intersection: line-convex polyhedron



Cyrus-Beck algorithm:

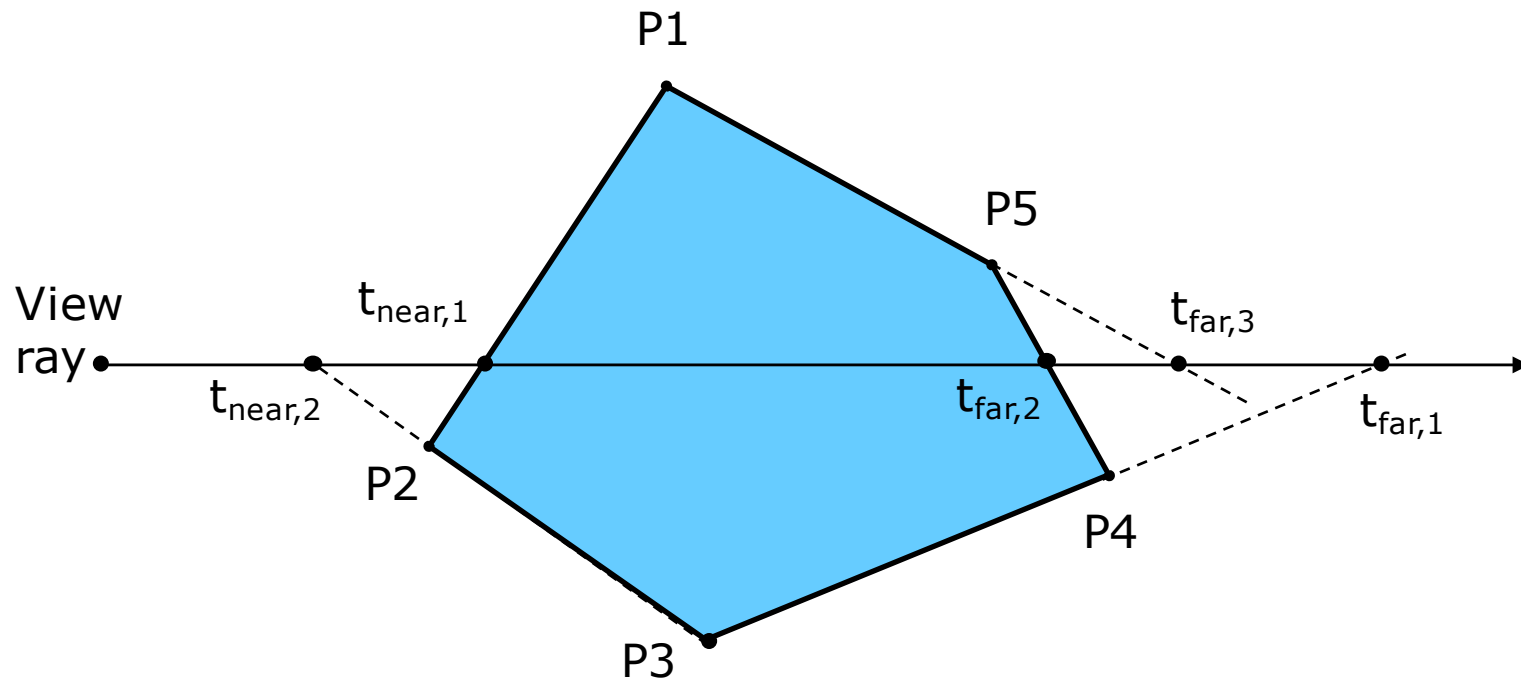
- Classify the edge/face by frontface and backface;

- For each edge/face compute intersection with view ray;

- Compute $\max(t_{near})$ and $\min(t_{far})$;

- If $\max(t_{near}) < \min(t_{far})$ there is intersection;

Intersection: line-convex polyhedron



Cyrus-Beck algorithm:

Initialize t_{near} to large negative value, t_{far} to large positive value

If (plane is back-facing) and ($t < t_{far}$) then $t_{far} = t$

If (plane is front-facing) and ($t > t_{near}$) then $t_{near} = t$

If ($t_{near} > t_{far}$) then (exit – ray misses)

Line and 3D object intersection

